

Headline-result evidence classes and replication units

 **FE Formal / executable**

 **SC Source-conditional**

 **CE Conditional empirical**

 **SI Scoped impl.**

 **O Open**

Nesting grammar — arrows mean contains, pairs, or repeats; use only row-declared levels



Result family + class	Estimand + unit	Declared unit (independent)	Nested structure	May support	Does not support
1 · Project identity at c=0 [FE]	paired outputs [categorical posterior]	exact identity	input → paired functions	project limit	full source protocol
2 · Client-loss robustness [SC]	generalized posterior [client update]	theorem; no empirical unit	assumptions → update → posterior	source theorem	server theorem; performance; calibration
3 · Paired free-energy contrast [CE]	isolated – shared colony F [nats]	configured seed	study → seed → condition → agents	paired protocol contrast	general benefit; exact replication
4 · Dirichlet likelihood learning [CE]	KL(true A learned A) [nats]	configured seed	study → seed → steps → A rows	executed conjugate path	universal rate; continuous model
5 · Configured BMR sign control [FE]	pruning delta F [nats]	closed-form comparison	posterior → pruning → signed delta	configured sign control	universal emergence
6 · Exploratory client baseline [CE]	joint-config accuracy [fraction]	synthetic seed	sweep → seed → client → row	composite configuration contrast	RCCE-only; calibration; BNN claims
7 · Server reweighting heuristic [CE]	robust – naive [p or fraction]	seed or matched trial	study → seed/world → trial → agents	finite mechanism behavior	influence bound; Byzantine tol.; universal win
8 · Variational server rule [FE]	descent, weights, transfer [mixed]	identity or seeded path	config → start → iterates	objective + tested transfer	global or universal claim
9 · Moving / disjoint-FOV contrasts [CE]	policy/sharing [fraction, nats, steps]	seed or fixed world	world → seed → condition → agents/steps	executed-world behavior	universal claim; physical multi-host
10 · Hierarchy / sensitivity diagnostics [CE]	posterior/gap/gate [probability, fraction, nats]	seed or fixed diagnostic	world → seed → level → agents/states	executed hierarchy design	universal benefit, depth, or emergence
11 · Acuity recovery diagnostic [CE]	acuity, error, R ² [p or unitless]	trial within acuity cell	grid point → trial → rows → estimate	finite-grid recovery	global structural identifiability
12 · Protocol-v1 / replay integrity [SI]	frame/replay match [frame or round]	local round	round → workers → frames → fields	protocol-v1 local integrity	encryption/mTLS; multi-host; Byzantine tol.
13 · Application receipt [SI]	hash/source agreement [artifact]	application invocation	invocation → artifacts → levels	hashes; requested source match	validity; decision; acceptance
14 · External calibration / confirmation [O]	paired log score [nats/observation]	dataset highest; seeds nested	dataset → seed → clients → rows	none before design + report	retrospective or pooled-dataset claim

- NT1 guarantees do not transfer between client, server, protocol, and application lanes
- NT2 nested agents, trials, and time steps are not promoted to independent datasets
- NT3 open external confirmation is not implied by formal or software evidence